

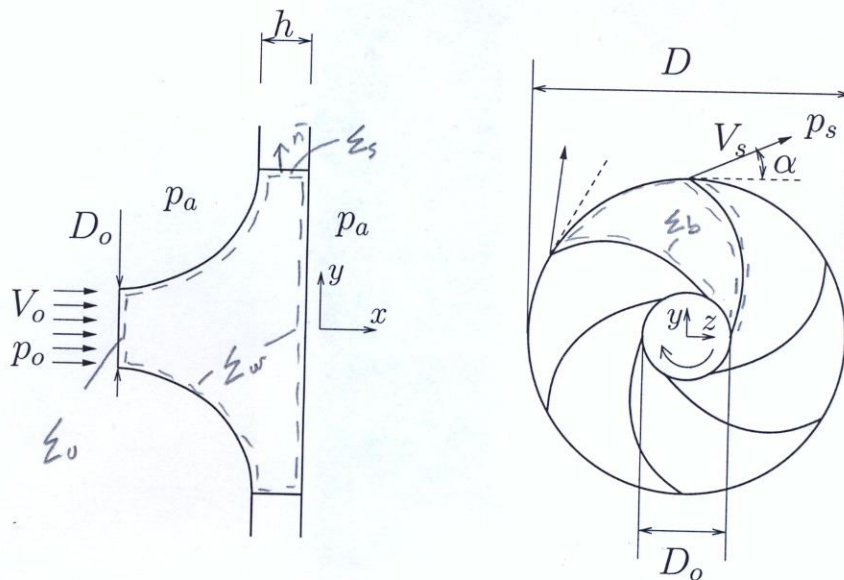
INTRODUCTION TO FLUID MECHANICS

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The figure shows a sketch of a centrifugal pump. Water enters the pump with uniform axial velocity V_o and pressure p_o through a circular pipe of diameter D_o . The internal geometry of the pump includes rotating blades that redirect the flow, so that the water leaves the pump with zero axial velocity through a lateral exit of width h and diameter D . At the exit section, the pressure is $p_s \neq p_a$ and the velocity vector, of magnitude V_s , forms an angle α with the lateral surface, as indicated in the figure, i.e., the corresponding normal and tangential velocity components are $V_s \sin \alpha$ and $V_s \cos \alpha$, respectively. Assuming that the effect of gravity is negligible and that the water temperature at the exit is equal to that found at the entrance determine the values of the

- Radial velocity component at the exit section $V_s \sin \alpha$.
- Axial force exerted on the pump.
- Torque exerted on the pump.
- Power of the pump, assumed to be ideal (i.e., work per unit time exerted on the water stream by the rotating parts).

→ THAT THE PUMP WALLS ARE INSULATED



Note: besides the water density ρ and the ambient pressure p_a , the following quantities can be assumed to be known in solving the problem: D , D_o , h , α , V_o , p_o , and p_s .

CONTINUITY: $\int_S (\vec{v} - \vec{v}_c) \cdot \vec{n} dS = 0 \Rightarrow -S V_o \pi \frac{D_o^2}{4} + S V_s \sin \alpha \pi D h = 0 \Rightarrow V_s \sin \alpha = \frac{D_o^2}{4 D h} V_o$

FORCE

$$\vec{F} = - \int_{S_{in} + S_b} p \vec{n} dS + \int_{S_{in} + S_b} \vec{z} \cdot \vec{n} dS - \int_{S_{out}} p_a \vec{n} dS = - \int_{S_{in} + S_b} (p - p_a) \vec{n} dS + \int_{S_{in} + S_b} \vec{z} \cdot \vec{n} dS$$

MOMENTUM

$$\int_S (\vec{v} - \vec{v}_c) \cdot \vec{n} dS = - \int_{S_{in} + S_b} p \vec{n} dS + \int_{S_{in} + S_b} \vec{z} \cdot \vec{n} dS \Rightarrow -S V_o \pi \frac{D_o^2}{4} V_o \vec{e}_x - (p_o - p_a) \pi \frac{D_o^2}{4} \vec{e}_x = - \int_{S_{in} + S_b} p \vec{n} dS + \int_{S_{in} + S_b} \vec{z} \cdot \vec{n} dS = -\vec{F}$$

TORQUE: APPLYING ANGULAR MOMENTUM WITH RESPECT TO THE ORIGIN

$$\int_S (\vec{r} - \vec{r}_o) \wedge S \vec{v} (\vec{v} - \vec{v}_c) \cdot \vec{n} dS = - \int_{S_{in}} (\vec{r} - \vec{r}_o) \wedge (p - p_a) \vec{n} dS + \int_{S_{in}} (\vec{r} - \vec{r}_o) \wedge \vec{z} \cdot \vec{n} dS$$

$$S V_o \pi \frac{D_o^2}{4} V_o \cos \alpha \frac{D}{2} \vec{e}_x = -\vec{M}$$

$$\vec{F} = \frac{\pi D_o^2}{4} \vec{e}_x [S V_o^2 + (p_o - p_a)]$$

ENERGY EQUATION

$$\int_S (\vec{e} + \frac{v^2}{2}) (\vec{v} - \vec{v}_c) \cdot \vec{n} dS = - \int_{S_{in}} p \vec{v} \cdot \vec{n} dS + \int_{S_{in}} \vec{v} \cdot \vec{z} \cdot \vec{n} dS + \int_{S_{in}} \kappa \vec{v} \cdot \vec{n} dS$$

$$S V_o \pi \frac{D_o^2}{4} \left(\frac{V_s^2}{2} - \frac{V_o^2}{2} \right) = + V_o \pi \frac{D_o^2}{4} (p_o - p_s) + W \Rightarrow W = S V_o \pi \frac{D_o^2}{4} \left[\left(\frac{p_s}{\rho} + \frac{V_s^2}{2} \right) - \left(\frac{p_o}{\rho} + \frac{V_o^2}{2} \right) \right]$$